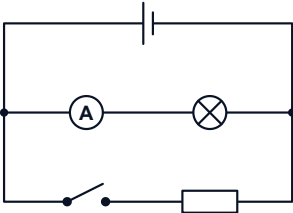


Series and parallel circuits

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

	QUESTION	ANSWER	✓
1	Name the two ways components can be joined in a circuit.	In series and in parallel	☐
2	Describe the current through components in series.	It is the same through each component.	☐
3	Describe the potential difference across components in series.	The total potential difference is shared between them.	☐
4	Describe the total resistance of components connected in series.	It is the sum of the individual resistances.	☐
5	Describe the current through components in parallel.	The total current is the sum of the currents in the branches.	☐
6	The lamp in this circuit is lit. Describe what happens to the ammeter reading when the switch is closed. 	It does not change. The lamp is on a separate branch, so the potential difference across it stays the same.	☐
7	Describe the potential difference across components in parallel.	It is the same across each component.	☐
8	Describe the total resistance of two resistors connected in parallel.	It is less than the resistance of the smaller resistor.	☐
9	Explain why adding resistors in series increases the total resistance.	The potential difference is shared between more resistors, so the current is smaller.	☐
10	Explain why adding resistors in parallel decreases the total resistance.	Each new branch gives the charge another path, so the total current increases.	☐

QUESTION		ANSWER	✓
11	Lamps are connected in series and one breaks. State what happens to the others.	They all go out, because the circuit is broken.	☐
12	Give two advantages of connecting lamps in parallel rather than in series.	Each lamp can be switched separately, and if one breaks the others stay lit.	☐